

Kappa electron energy distributions in Mappings

Mappings allows the use of non-equilibrium electron energy distributions (described by the kappa parameter) in photoionization calculations.

Conventionally, when modelling emission nebulae we assume that the electrons are in thermal equilibrium, with energies described by the Maxwell-Boltzmann distribution. However, satellite and space probe direct measurements of electron energies in the solar system find non-equilibrium energy distributions are the norm. Many of these can be described by the "kappa" distribution. Such distributions may also occur in nebulae, under similar physical conditions.

Kappa distributions for the nebular electron energies also provide a simple explanation for a long standing problem where measurements of electron temperatures and chemical abundances using collisionally excited and optical recombination emission lines yield discrepant results. This has been a major concern, as measurements of nebular abundances are widely used to study galaxy evolution and dynamics.

The effect of a kappa distribution is to enhance lines at low energies (typical of recombination lines) and to enhance lines arising from higher energy levels (eg the 1S_0 level of [O III] which gives rise to the 4363Å line), and to diminish lines arising from intermediate energy levels (eg the 1D_2 level of [O III] which gives rise to the 5007Å line). This can affect in different ways, for example, the "direct method" temperatures calculated for different ionic species (in addition to any intrinsic geometry variation in these values).

To investigate this, the option to calculate nebular photoionization models using kappa electron energy distributions was incorporated into Mappings.

See Nicholls et al. (2012, 2013, 2017 and references therein) (NASA ADS 2012ApJ...752..148N, 2013ApJS..207...21N, and 2017mcp..book...633N) for a thorough discussion, and the *Background* section below for a summary of the arguments.

Computing spectra with kappa energy distribution electrons

This option is not available by default in the Mappings command sequence. It can be activated as a menu item by editing the file *switches.txt* located in the *addons* directory, and setting the kappa option to 1 from the default zero, then copying the edited file into the Mappings *data* directory.

Once this is done, the third command option after running the Mappings executable will be "Use Kappa electron distributions? (y/N)"

Answering this as "y" then generates the prompt to set the value of kappa. Values between 2 and 1000 are allowed (kappa = infinity corresponds to the standard Maxwell-Boltzmann distribution). Choose a suitable value -- the lower the value, the greater the departure of the distribution from the M-B distribution.

The computation then proceeds as for the previous default options.

Background to non-Maxwellian electron energy distributions

It has been accepted since the 1940s that the electrons in HII regions and PNe are in thermal equilibrium. Analytical calculations of electron velocity distributions in gaseous nebulae were presented by Bohm & Aller (1947, ADS 1947ApJ...105..131B). They reported that the velocity distribution is "very close to Maxwellian." Spitzer (1962, Ch. 5, ADS 1968dms..book.....S) also examined the thermalization process for electron energies in plasmas and found that electron energies equilibrate rapidly through collisions. This early work has led later authors to assume that the electrons in gaseous nebulae are always in thermal equilibrium. However, Spitzer's analysis showed that the equilibration time of an energetic electron is proportional to the cube of the velocity, so using M-B equilibrium theory, plasmas with very high energy electrons take much longer to equilibrate than those excited by normal UV photons from stars found in HII regions.

In more recent times, the electron energies in solar system plasmas have been measured directly by satellites and space probes, and in most cases the electron energies depart substantially from the Maxwellian, and resemble a Maxwellian with a high energy power law tail. Vasyliunas (1968, ADS 1968JGR....73.2839V) showed that this distribution could be well described by what he called the "kappa distribution". There is also evidence from IBEX observations that energetic neutral atoms in the interstellar medium, where it interacts with the heliosheath, exhibit kappa energy distributions. In solar system plasmas, the kappa distribution is more common than the M-B distribution. So we are confronted with the fact that despite the early theoretical work suggesting that the electrons in such plasmas should be in thermal equilibrium, frequently they are not.

The kappa distribution has been shown to arise naturally from "q non-extensive statistical mechanics", in the same way as the M-B distribution arises from Boltzmann-Gibbs statistics. The requirement for this to occur is that there be macroscopic interactions between forces and particles, in addition to the shorter-range Coulombic forces that give rise to M-B equilibration. These statistics provide a sound basis for the overtly successful use of the kappa distribution in describing solar system plasmas.

Kappa distributions appear to arise whenever the plasma is being pumped rapidly with high energy non-thermal electrons, so that the system cannot relax to a classical M-B distribution. Kappa-like energy distributions can also arise as a consequence of normal power-law variations of physical parameters such as density, temperature, and electric and magnetic fields.

Such conditions may also be present in HII regions and PNe—solar system plasma parameters span the many of the conditions found in gaseous nebulae, and, as in the solar system, HII region plasmas can be magnetically dominated—so it is important to investigate the potential effects of non-equilibrium energy distributions in gaseous nebulae.

Such non-Maxwellian energies may occur whenever the population of energetic electrons is being pumped in a timescale $<$ the normal energy redistribution timescale of the electron population. Suitable mechanisms include magnetic reconnection followed by the migration of high-energy electrons along field lines, the development of inertial Alfvén waves, local shocks (driven either by the collision of bulk flows or by supersonic turbulence), and, most simply, by the injection of high-energy electrons through the photoionization process itself. Normal photoionization produces supra-thermal electrons on a timescale similar to the recombination timescale. However, energetic electrons can be generated by the photoionization of dust, and X-ray ionization can produce highly energetic ($\sim$ keV) inner-shell (Auger process) electrons. These photoionization-based processes are more effective where the source of the ionizing photons has a “hard” photon spectrum. Thus, the likelihood of the ionized plasma having a kappa electron energy distribution would be high in the case of either photoionization by an active galactic nucleus, in PNe, where the effective temperature of the exciting star can range up to $\sim 250,000$ K, and for high-redshift HII regions excited by very low metallicity O-stars.

So we have no shortage of possible energy injection mechanisms capable of feeding the energetic population on a timescale which is short compared with the collisional re-distribution timescale. The rate of equilibration falls rapidly with increasing energy, and we would expect there to be a threshold energy above which any non-thermal electrons have a long residence time. These can then feed continually down toward lower energies through conventional collisional energy redistribution, thus maintaining a kappa electron energy distribution.

In addition to the energy injection mechanisms capable of maintaining the excitation of supra-thermal distributions, there is evidence that the kappa distribution may remain stable against equilibration longer than expected through simple collisional thermalization, and that distributions with $2.5 > \kappa > 1.5$ have the capacity, through increasing entropy, to move to values

of lower kappa i.e., away from (M-B) equilibrium. This suggests that where q non-extensive entropy conditions operate, the supra-thermal energy distributions produced exist in time-invariant “stationary states”. These states have longer lifetimes than expected classically. This is consistent with the numerous observations in solar system plasmas that kappa electron and proton energy distributions are the norm.

In addition, for several decades, systematic discrepancies have plagued abundance measurements derived from observations of emission lines and emission continua in HII regions and PNe. In particular, abundances determined from collisionally excited lines (CELs) differ from those determined using hydrogen and helium bound-free continuum spectra. Chemical abundances determined from the optical recombination lines (ORLs) are systematically higher than those determined from CELs, these so-called “abundance discrepancy problem”. The kappa electron energy distribution provides a simple explanation for why this happens.

It is likely, therefore, that photoionized plasmas in gaseous nebulae will show departures from a Maxwell distribution to some degree. The key questions are, is this important, and does it produce observable effects in the nebular diagnostics which we have relied upon hitherto?

The best way to answer these questions is to model nebulae (HII regions, AGNs, PNe and shocks) using kappa-distributed electron energy models to compare with observations. For this reason, Mappings has the capability to model emission lines and continua using kappa-distributed electron energies.